

3. Using the MNIST dataset (digit vs not digit), implement a binary classifier.**CODE:**

```
import pandas as pd
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load Dataset
digits = load_digits()
print("Dataset Loaded.")

# Features and Target
X = digits.data
Y = digits.target

# Binary Classification (True if digit is 5)
Y = (Y == 5)

# Split Dataset
X_Train, X_Test, Y_Train, Y_Test = train_test_split(
    X, Y, test_size=0.3, random_state=1
)

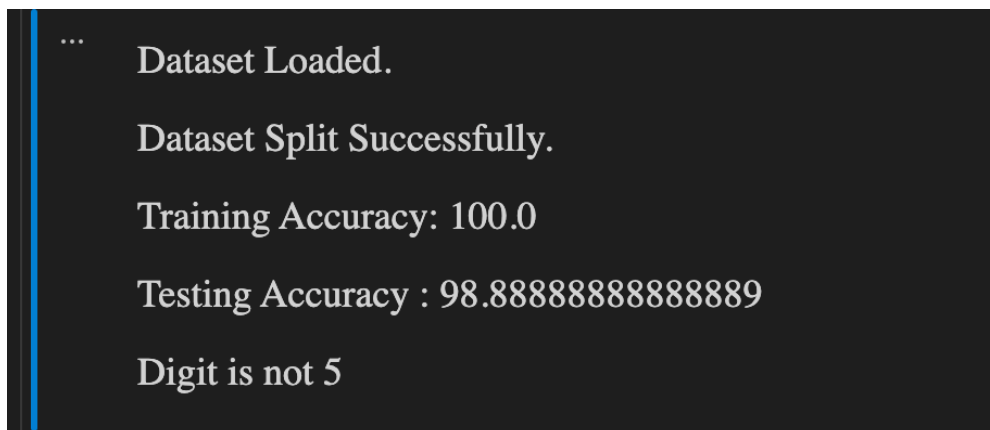
print("Dataset Split Successfully.")

# Train Model
model = LogisticRegression(max_iter=1000)
model.fit(X_Train, Y_Train)

# Accuracy
print("Training Accuracy:", model.score(X_Train, Y_Train) * 100)
print("Testing Accuracy :", model.score(X_Test, Y_Test) * 100)

# Predict First Test Sample
prediction = model.predict([X_Test[0]])

if prediction[0]:
    print("Digit is 5")
else:
    print("Digit is not 5")
```

OUTPUT:A screenshot of a terminal window with a dark background and light blue text. The output shows the execution of the code, including dataset loading, splitting, training, and prediction results. The prediction for the first test sample is 'Digit is not 5'.

```
... Dataset Loaded.

Dataset Split Successfully.

Training Accuracy: 100.0

Testing Accuracy : 98.88888888888889

Digit is not 5
```

3.1 Modify the program to detect Digit 7 vs Not Digit 7 instead of Digit 5 vs Not Digit 5.**CODE:**

```
import pandas as pd
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load Dataset
data = pd.read_csv("mnist_test.csv") # Loaded but not used
digits = load_digits()

# Features and Target
X = digits.data
y = digits.target

# Binary Classification (Digit = 7)
y = (y == 7)

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

# Train Model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# Accuracy
print("Training Accuracy :", model.score(X_train, y_train) * 100)
print("Testing Accuracy :", model.score(X_test, y_test) * 100)

# Predict First Test Sample
prediction = model.predict([X_test[0]])

if prediction[0]:
    print("Digit is 7")
else:
    print("Digit is not 7")
```

OUTPUT:

Training Accuracy : 100.0

Testing Accuracy : 99.07407407407408

Digit is not 7

3.2 Modify the program to detect Digit 0 vs Not Digit 0.**CODE:**

```
import pandas as pd
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load Dataset
data = pd.read_csv("mnist_test.csv") # Loaded but not used
digits = load_digits()

# Features and Target
X = digits.data
y = digits.target

# Binary Classification (True if digit is 0)
y = (y == 0)

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

# Train Model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# Accuracy
print("Training Accuracy :", model.score(X_train, y_train) * 100)
print("Testing Accuracy :", model.score(X_test, y_test) * 100)

# Predict First Test Sample
prediction = model.predict([X_test[0]])

if prediction[0]:
    print("Digit is 0")
else:
    print("Digit is not 0")
```

OUTPUT:

Training Accuracy : 100.0

Testing Accuracy : 99.44444444444444

Digit is not 0

3.3. Modify the program to detect Digit 9 vs Not Digit 9.**CODE:**

```
import pandas as pd
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load Dataset
data = pd.read_csv("mnist_test.csv") # Loaded but not used
digits = load_digits()

# Features and Target
X = digits.data
y = digits.target

# Binary Classification (True if digit is 9)
y = (y == 9)

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

# Train Model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# Accuracy
print("Training Accuracy :", model.score(X_train, y_train) * 100)
print("Testing Accuracy :", model.score(X_test, y_test) * 100)

# Predict First Test Sample
prediction = model.predict([X_test[0]])

if prediction[0]:
    print("Digit is 9")
else:
    print("Digit is not 9")
```

OUTPUT:

Training Accuracy : 99.68178202068417

Testing Accuracy : 98.51851851851852

Digit is not 9

3.4. Modify the program to display the first 20 predictions instead of only one prediction.

CODE:

```
import pandas as pd
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load Dataset
data = pd.read_csv("mnist_test.csv") # Loaded but not used
digits = load_digits()

# Features and Target
X = digits.data
y = digits.target

# Binary Classification (True if digit is 9)
y = (y == 9)

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

# Train Model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# Accuracy
print("Training Accuracy :", model.score(X_train, y_train) * 100)
print("Testing Accuracy :", model.score(X_test, y_test) * 100)

# Predict First 20 Test Samples
predictions = model.predict(X_test[:20])

for prediction in predictions:
    if prediction:
        print("Digit is 9")
    else:
        print("Digit is not 9")
```

OUTPUT:

```
Training Accuracy : 99.68178202068417
Testing Accuracy : 98.51851851851852
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
```

```
Digit is 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is not 9
Digit is 9
Digit is not 9
Digit is not 9
Digit is not 9
```